

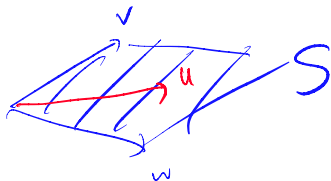
Linear Algebra Recitation

Week 6

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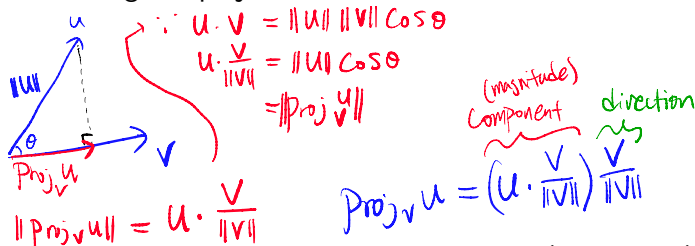
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(Orthogonal) Projection matrices

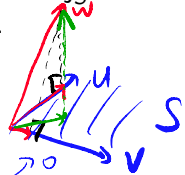
- $\{u_1, \dots, u_n\}$ is an orthogonal set of vectors if: $u_i \cdot u_j = 0 \quad i \neq j$
- $\{u_1, \dots, u_n\}$ is an orthonormal set of vectors if: $\downarrow + \|u_i\| = 1$
- V is a finite dimensional vector space, and let $u, v \in V$. Recall the orthogonal projection of a vector v to a vector u .



- In general, a real square matrix P is an (orthogonal) projection matrix iff $P^2 = P$ and $P^T = P$. $PP^T = P^T P = P$
- Note: We are mostly concerned about orthogonal projections in this course, thus a "projection" will refer to an orthogonal projection.

(Orthogonal) Projection matrices

- Projection of a vector to higher dimensions.
- Let $u, v, w \in \mathbb{R}^3$, and u, v are orthogonal. Define $S = \text{span}(u, v)$ (hence u, v form an orthogonal basis for S). Find $\text{Proj}_S w$. Also, find the matrix P such that $Pw = \text{Proj}_S w$.


$$\begin{aligned}\text{Proj}_S w &= c_1 u + c_2 v \\ &= \left(w \cdot \frac{u}{\|u\|} \right) \frac{u}{\|u\|} + \left(w \cdot \frac{v}{\|v\|} \right) \frac{v}{\|v\|} \\ (w - \text{Proj}_S w) \cdot u &= w \cdot u - \left(w \cdot \frac{u}{\|u\|} \right) \frac{u \cdot u}{\|u\|} - \left(w \cdot \frac{v}{\|v\|} \right) \frac{v \cdot u}{\|v\|} \\ &= w \cdot u - \left(w \cdot \frac{u}{\|u\|} \right) \|u\| = w \cdot u - w \cdot u = 0\end{aligned}$$

- An orthogonal projection means that $w - \text{Proj}_S w$ is orthogonal to every vector in S , or equivalently, $w - \text{Proj}_S w \in S^\perp$.

(Orthogonal) Projection matrices

- Generalization: Let $w, u_1, \dots, u_k \in V$, where $u_1, \dots, u_k$ are orthogonal. Define $S = \text{span}(u_1, \dots, u_k)$. Find P such that

$$\|u_i\| = 1 \\ \forall i=1, \dots, k \\ u_i \in \mathbb{R}^n$$

$$Pw = \text{Proj}_S w = \left\| \text{Proj}_{u_1} w \right\| \frac{u_1}{\|u_1\|} + \dots + \left\| \text{Proj}_{u_k} w \right\| \frac{u_k}{\|u_k\|}$$

$$= (w \cdot u_1) u_1 + \dots + (w \cdot u_k) u_k$$

$$U U^T w = [u_1 \dots u_k] \begin{bmatrix} u_1 \cdot w \\ \vdots \\ u_k \cdot w \end{bmatrix}$$

$$U = [u_1 | \dots | u_k]$$

$$P = U U^T$$

$$= \begin{bmatrix} u_{11}(u_1 \cdot w) + u_{21}(u_2 \cdot w) + \dots + u_{k1}(u_k \cdot w) \\ \vdots \\ u_{1n}(u_1 \cdot w) + \dots + u_{kn}(u_k \cdot w) \end{bmatrix}$$

$$U^T = \begin{bmatrix} u_1^T \\ \vdots \\ u_k^T \end{bmatrix}$$

$$U^T w = \begin{bmatrix} u_1^T w \\ \vdots \\ u_k^T w \end{bmatrix} = \begin{bmatrix} u_1 \cdot w \\ \vdots \\ u_k \cdot w \end{bmatrix}$$

$$= (u_1 \cdot w) \begin{bmatrix} u_{11} \\ \vdots \\ u_{1n} \\ \vdots \\ u_{k1} \\ \vdots \\ u_{kn} \end{bmatrix} + \dots + (u_k \cdot w) \begin{bmatrix} u_{k1} \\ \vdots \\ u_{kn} \end{bmatrix} = (u_1 \cdot w) u_1 + \dots + (u_k \cdot w) u_k$$

- We have only considered projecting on a space spanned by an orthogonal basis. This is because any basis can be turned into an orthonormal basis via Gram-Schmidt process.

When $\{u_1, \dots, u_k\}$ orthonormal

$$\Rightarrow \underline{P = UU^T}$$

If not Orthonormal, but still orthogonal

$$\text{Proj}_S w = (w \cdot \frac{u_1}{\|u_1\|}) \frac{u_1}{\|u_1\|} + \dots + (w \cdot \frac{u_k}{\|u_k\|}) \frac{u_k}{\|u_k\|}$$

$$U = [u_1 \dots u_k], \quad U^T = \begin{bmatrix} \frac{u_1^T}{\|u_1\|} \\ \vdots \\ \frac{u_k^T}{\|u_k\|} \end{bmatrix}$$

$$U^T U = \begin{bmatrix} u_1^T u_1 & & 0 \\ & \ddots & \\ 0 & & u_k^T u_k \end{bmatrix}$$

$$\underline{(U^T U)^{-1}} = \begin{bmatrix} \frac{1}{\|u_1\|^2} & & \\ & \ddots & \\ & & \frac{1}{\|u_k\|^2} \end{bmatrix}$$

$$\underline{P = U(U^T U)^{-1} U^T}$$

$$P^2 = U \cancel{(U^T U)^{-1}} \cancel{(U^T U)} (U^T U)^{-1} U^T \\ = U(U^T U)^{-1} U^T = P$$

if $\{u_1, \dots, u_k\}$ not even orthogonal

$\Rightarrow$ the dot product formula does not work, and deriving the formula directly is more complicated.

instead,

some projection onto S

We want to find $z = c_1 u_1 + \dots + c_k u_k = Uc$ such that $\|w - z\| = \|w - Uc\|$ is minimized (orthogonal projection)

this is called a least-squares problem, and can be solved by solving $U^T U c = U^T w$ (normal equation).

If $\{u_1, \dots, u_k\}$ are at least linearly independent, then $c = (U^T U)^{-1} U^T w$ is the unique minimizer of $\|w - Uc\|$. (can be shown via simple calculus).

Therefore, the orthogonal projection of w onto S is

$z = Uc = U(U^T U)^{-1} U^T w$, and $P = U(U^T U)^{-1} U^T$

(Orthogonal) Projection matrices

- Exercise: Let $A = \begin{bmatrix} 1 & 2 \\ 0 & -1 \\ -1 & 0 \end{bmatrix}$ and $b = \begin{bmatrix} 0 \\ 1 \\ -1 \end{bmatrix}$. Find $\text{Proj}_{\text{col}(A)} b$.

$\{u_1, u_2\}$

$$\underset{\text{span}(u_1, u_2)}{\text{Proj}_{\text{col}(A)}} b = (b \cdot u_1) u_1 + (b \cdot u_2) u_2$$

or

$$U = [u_1 \mid u_2]$$

$$P = U U^T$$

$$\text{Proj}_{\text{col}(A)} b = P b$$

Orthonormal Basis of a Vector Space

- An orthonormal basis is (i) a basis, (ii) that is an orthonormal set.
- Demo of why a orthonormal basis is desirable:

Let $u, v, w \in \mathbb{R}^3$, and u, v form an orthonormal set. Define $S = \text{span}(u, v)$ (hence u, v form a orthonormal basis for S).

Compute $\|\text{Proj}_u w\|u + \|\text{Proj}_v w\|v = \text{Proj}_u w + \text{Proj}_v w$

$$= w \cdot \frac{u}{\|u\|} = w \cdot u$$

$$[w - (\text{Proj}_u w + \text{Proj}_v w)] \in S^\perp$$

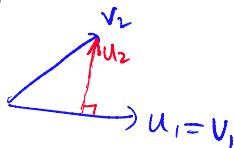
(only because u, v orthogonal)

- What if $\{u, v\}$ is a basis of S that is not orthonormal?
 $\implies \|\text{Proj}_u w\| \frac{u}{\|u\|} + \|\text{Proj}_v w\| \frac{v}{\|v\|}$ may **not** be an orthogonal projection of w to S .

The Gram-Schmidt process

- Every basis can be turned into an orthonormal basis.
- Input: a basis $\{v_1, \dots, v_n\}$ of $V \longrightarrow$ Output: an **orthonormal** basis $\{u_1, \dots, u_n\}$ of V .
- Step 1: Start with $u_1 = v_1$.

$$u_2 = v_2 - \text{Proj}_{u_1} v_2, \quad u_2 = \frac{u_2}{\|u_2\|}$$



$$u_3 = v_3 - \text{Proj}_{\text{span}(u_1, u_2)} v_3 = v_3 - \left[\|\text{Proj}_{u_1} v_3\| u_1 + \|\text{Proj}_{u_2} v_3\| u_2 \right]$$

$$u_4 = v_4 - \text{Proj}_{\text{span}(u_1, u_2, u_3)} v_4$$

$$u_n = v_n - \text{Proj}_{\text{span}(u_1, \dots, u_{n-1})} v_n$$

The Gram-Schmidt process

$$\begin{aligned}\|v_1\| &= \sqrt{10} \\ \|v_2\| &= \sqrt{5} \\ \|v_3\| &= \sqrt{11}\end{aligned}$$

- Exercise: $A = \begin{bmatrix} v_1 & v_2 & v_3 \\ 1 & 2 & 1 \\ 2 & -1 & 3 \\ 2 & 3 & 0 \\ 1 & -1 & 1 \end{bmatrix}$. Find an orthonormal basis for $\text{col}(A)$.

$$u_1 = v_1 / \|v_1\| = v_1 / \sqrt{10}$$

$$u_2 = v_2 - (v_2 \cdot u_1) u_1 = \begin{bmatrix} 2 \\ -1 \\ 3 \end{bmatrix} - \frac{15}{2\sqrt{10}} \frac{1}{\sqrt{10}} \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix} = \begin{bmatrix} 3/2 \\ -2 \\ 2 \\ -3/2 \end{bmatrix} \rightarrow \text{normalize}$$

$$u_3 = v_3 - (v_3 \cdot u_1) u_1 - (v_3 \cdot u_2) u_2 = \text{DIY}$$